Centre of Experimental Physics,
Harbin Institute of Technology, Shenzhen

University Physics Experiments 1A (PHYS1002AE)

Introductory Lecture @ T3501, 18:45 - 21:35, 3rd March 2025

Ho-Kei Chan (陈浩基)
School of Science,
Harbin Institute of Technology, Shenzhen

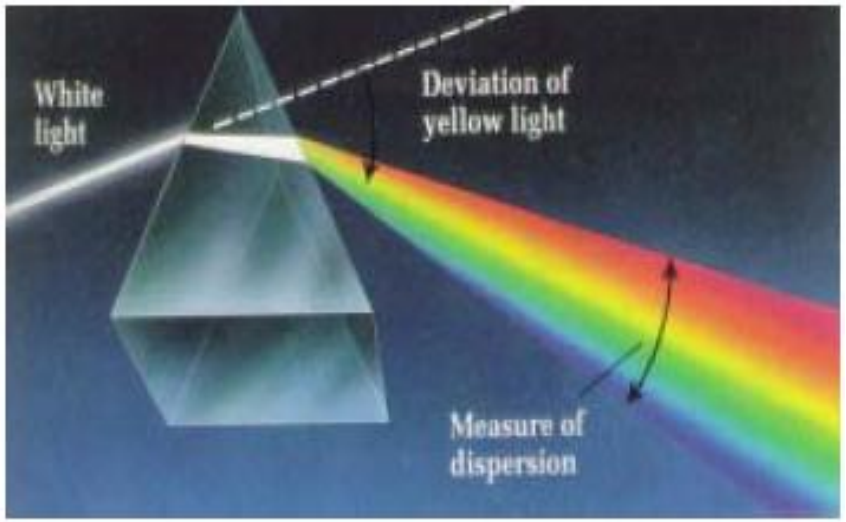
Email: hokeichan@hit.edu.cn

Outline

1. Course background
2. Course objectives
3. Course contents
4. Course arrangements
5. Course requirements
6. Errors and uncertainties in measurements
7. Significant figures of measured quantities
8. Methods of data analysis

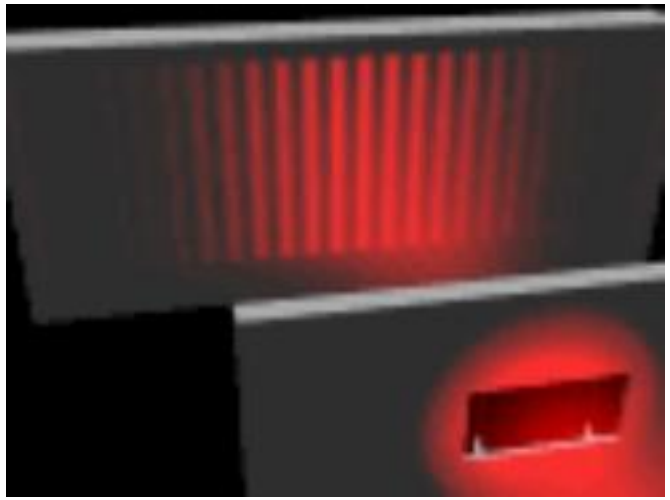
1. Course background

Experimental physics focuses on the acquisition and analysis of empirical data, with the aim to advance our knowledge and understanding of the physical world by confirming or disproving existing theories of physics. Here are some classic examples:

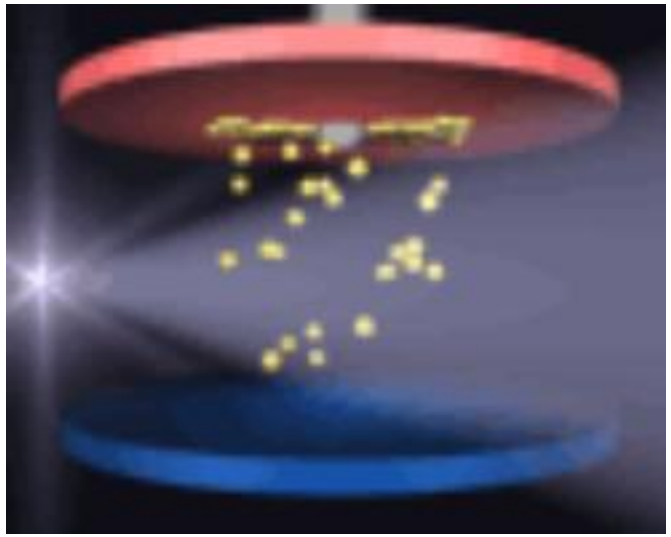


Newton's decomposition of sunlight with a prism (1665-1666), confirming that white light is composed of seven visible colors.

1. Course background

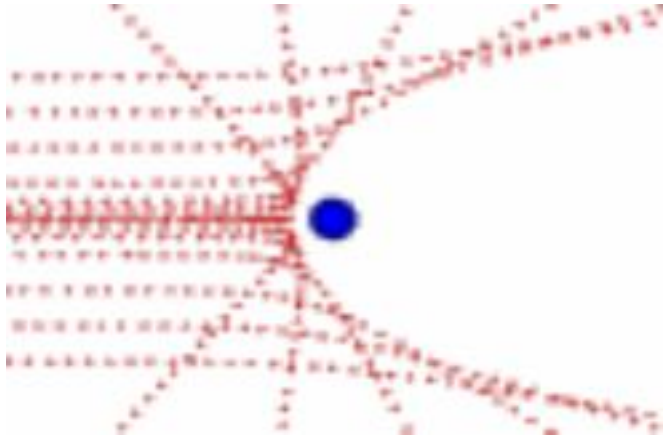


Young's light-interference experiment (1801), confirming the wave nature of light.

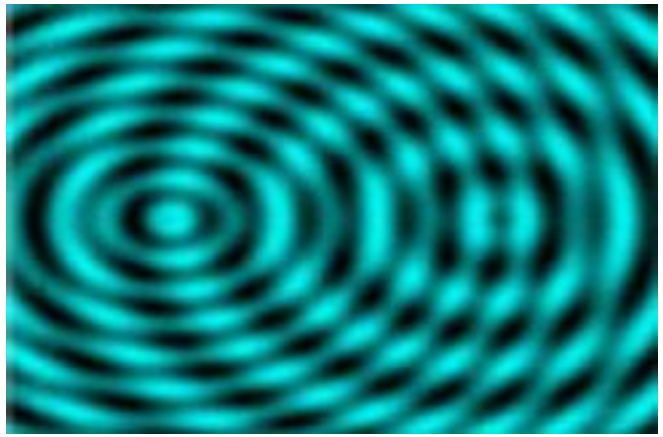


Millikan's oil-drop experiment (1909), obtaining an accurate value for the elementary electric charge carried by an electron.

1. Course background



Rutherford's gold foil experiment (1911), resulting in the discovery of the nucleus.



Young's double-slit experiment applied to the interference of single electrons (1961), confirming the wave nature of electrons.

2. Course objectives

"University Physics Experiments" is an compulsory course for all STEM majors across the university. It offers systematic training of experimental skills and experimental methodology for undergraduate students, with the following objectives:

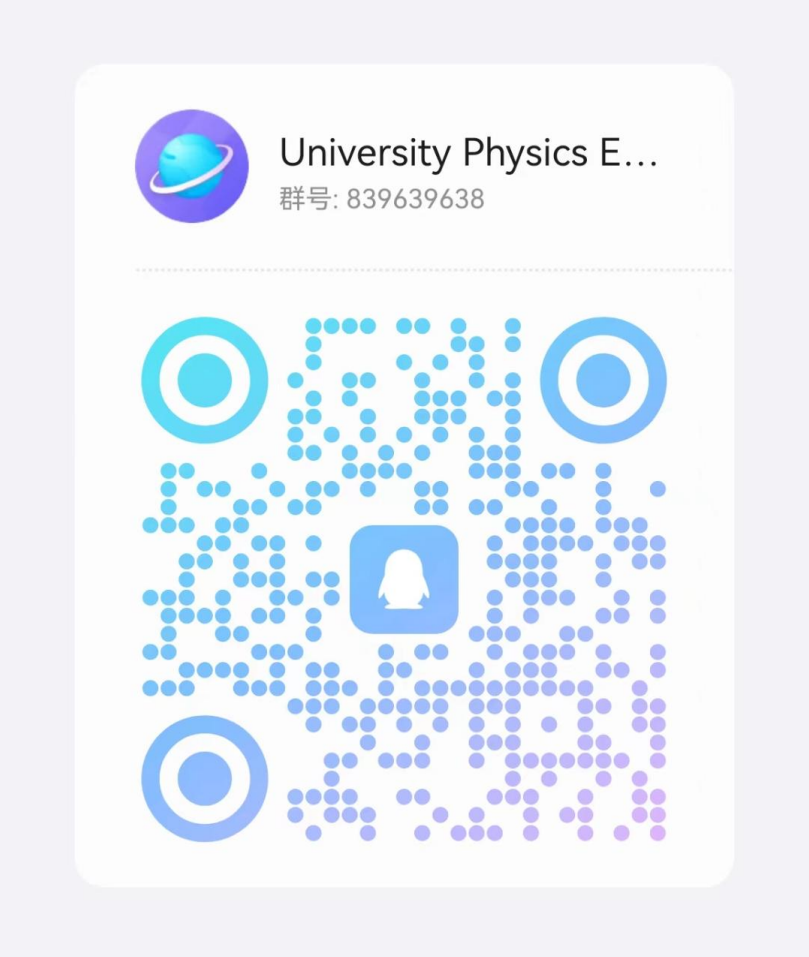
1. To equip students with basic experimental skills, a scientific mindset, and the ability to innovate.
2. To equip students with the ability to properly acquire, record and analyse experimental data using existing knowledge of physics.

3. Course contents - 1A, spring semester, 2025)

Experiment	Laboratory	Lecturer
I. Michelson interferometer	T5707	Rui ZHONG (钟瑞)
II. Young's Modulus of a Metallic Wire	T5808	Feng CAO (曹峰) / Sisi HE (何思斯)
III. Collision and Targeting Experiment	T5706	Boyi WANG (王博弋)/ Weijian ZHOU (周伟健)
IV. The Hall Effect	T5802	Fei YEN (晏飞翔)
V. Measurement of the Surface-Tension Coefficient of a Liquid	T5801	Yanshi HUANG (黄延实)
VI. Characteristics of RLC circuits	T2707	Xiaobin CHEN (陈晓彬)/ Weijian ZHOU (周伟健)
VII. Speed of Sound in Air	T5805	Ho-Kei CHAN (陈浩基) / Sisi HE (何思斯)

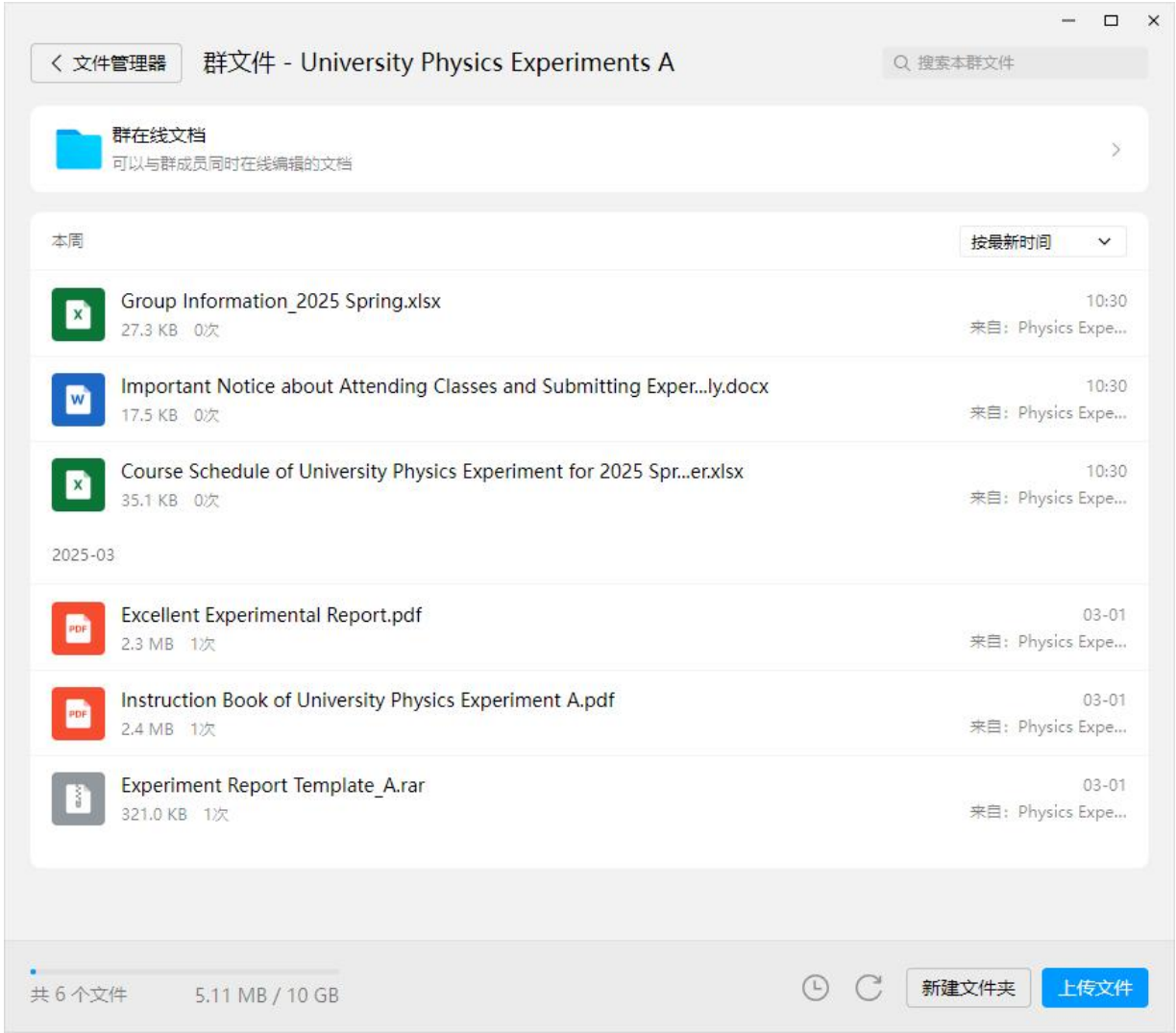
4. Course arrangements

QQ group (managed by Rui ZHONG)



4. Course arrangements

Important documents which you can download after joining the QQ group



4. Course arrangements - schedule for week 3 to week 16

Course Schedule of University Physics Experiment for 2025 Spring Semester

Note: Some classes start at **13:15** (Because the Department of Academic Affairs has not reserved enough time for class scheduling), be c

Week	Day	Date	Class	Time	Only for University Physics Experiment IA	For Both University Physics Experiment IA and IB					
					Michelson Interferometer (Room:T5707) Instructor: 钟瑞	Speed of Sound in Air (Room:T5805) Instructor: 陈浩基，何思斯	Characteristics of RLC Circuits (Room:T2707) Instructor: 陈晓彬，周伟建	Hall Effect (Room:T5802) Instructor: 晏飞翔	Young's Modulus of a Metallic Wire (Room:T5808) Instructor: 曹峰，何思斯	Collision and Targeting Experiment (Room:T5706) Instructor: 王博弋，周伟建	Measurement of the Surface-tension Coefficient of a Liquid (Room:T5801) Instructor: 黄延实
3	Monday	2025. 03. 10	Noon-5-6	13:15-15:45		A8		A12			
			5-6-7	14:00-16:30							
	Tuesday	2025. 03. 11	7-8-Nightfall	16:00-18:30							
	Friday	2025. 03. 14	Noon-5-6	13:15-15:45							
4	Monday	2025. 03. 17	Noon-5-6	13:15-15:45	A7		A9		A10	A11	A12
			5-6-7	14:00-16:30							
	Tuesday	2025. 03. 18	7-8-Nightfall	16:00-18:30							
	Friday	2025. 03. 21	Noon-5-6	13:15-15:45							
5	Monday	2025. 03. 24	Noon-5-6	13:15-15:45	A12	A7	A8		A9	A10	A11
			5-6-7	14:00-16:30							
	Tuesday	2025. 03. 25	7-8-Nightfall	16:00-18:30							A1
	Friday	2025. 03. 28	Noon-5-6	13:15-15:45							

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6	Monday	2025. 03. 31	Noon-5-6	13:15-15:45	A11	A12	A7		A8	A9	
			5-6-7	14:00-16:30							
	Tuesday	2025. 04. 01	7-8-Nightfall	16:00-18:30							A2
7	Monday	2025. 04. 07	Noon-5-6	13:15-15:45	A9			A10		A7	
			5-6-7	14:00-16:30		B1			B3		
	Tuesday	2025. 04. 08	7-8-Nightfall	16:00-18:30	A1	A2	A3	A4	A5	A6	
	Friday	2025. 04. 11	Noon-5-6	13:15-15:45	A13	A14	A15	A16			
8	Monday	2025. 04. 14	Noon-5-6	13:15-15:45	A8	A9		A11			A7
			5-6-7	14:00-16:30			B1		B2	B3	
	Tuesday	2025. 04. 15	7-8-Nightfall	16:00-18:30	A2	A3	A4	A5	A6	A1	
	Friday	2025. 04. 18	Noon-5-6	13:15-15:45	A14	A15	A16	A13			

4. Course arrangements - schedule for week 3 to week 16

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9	Monday	2025. 04. 21	Noon-5-6	13:15-15:45	A10	A11	A12	A7			
			5-6-7	14:00-16:30					B1	B2	B3
	Tuesday	2025. 04. 22	7-8-Nightfall	16:00-18:30	A3	A4	A5	A6	A1	A2	
	Friday	2025. 04. 25	Noon-5-6	13:15-15:45	A15	A16	A13	A14			
10	Monday	2025. 04. 28	Noon-5-6	13:15-15:45		A10	A11		A12	A8	
			5-6-7	14:00-16:30				B1			B2
	Tuesday	2025. 04. 29	7-8-Nightfall	16:00-18:30	A4	A5	A6	A1	A2	A3	
11	Tuesday	2025. 05. 06	7-8-Nightfall	16:00-18:30	A5	A6	A1	A2	A3	A4	
	Friday	2025. 05. 09	Noon-5-6	13:15-15:45	A16	A13	A14	A15			
12	Monday	2025. 05. 12	Noon-5-6	13:15-15:45			A10		A7	A12	
			5-6-7	14:00-16:30				B3			B1
	Tuesday	2025. 05. 13	7-8-Nightfall	16:00-18:30	A6	A1	A2	A3	A4	A5	
	Friday	2025. 05. 16	Noon-5-6	13:15-15:45					A13	A14	A15

4. Course arrangements - schedule for week 3 to week 16

Course Schedule of University Physics Experiment for 2025 Spring Semester

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13	Monday	2025. 05. 19	Noon-5-6	13:15-15:45					A11		A9
			5-6-7	14:00-16:30				B2			
	Tuesday	2025. 05. 20	7-8-Nightfall	16:00-18:30							A3
	Friday	2025. 05. 23	Noon-5-6	13:15-15:45					A14	A15	A16
14	Monday	2025. 05. 26	Noon-5-6	13:15-15:45				A8			A10
			5-6-7	14:00-16:30		B2	B3				
	Tuesday	2025. 05. 27	7-8-Nightfall	16:00-18:30							A4
	Friday	2025. 05. 30	Noon-5-6	13:15-15:45					A15	A16	A13
15	Tuesday	2025. 06. 03	7-8-Nightfall	16:00-18:30							A5
	Friday	2025. 06. 06	Noon-5-6	13:15-15:45					A16	A13	A14
16	Monday	2025. 06. 09	Noon-5-6	13:15-15:45				A9			A8
			5-6-7	14:00-16:30		B3	B2			B1	
	Tuesday	2025. 06. 10	7-8-Nightfall	16:00-18:30							A6
	Friday	2025. 06. 13	Noon-5-6	13:15-15:45							

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
1	2024331010	MOHSEN KHALED QAIDA BDULLAH	24级计算机（留学生）1班	A1
2	2024331015	ALMABR ABDULLAH FAREA MOHAMMED AHMED	24级计算机（留学生）1班	
3	2024331016	ASSOUALEM YOUSSEF	24级计算机（留学生）1班	
4	2024331029	CHOURI YASSINE	24级计算机（留学生）1班	
5	2024331030	AMGALANBAATAR ERDENE	24级计算机（留学生）1班	
6	2024331037	RAIMBEKOVA INARA	24级计算机（留学生）1班	
7	2024331046	BENDAOUD RACHID	24级计算机（留学生）1班	
8	2024331048	SHATKOV TIMOFEI	24级计算机（留学生）1班	
9	2024331053	BAT-ERDENE BOLD-ERDENE	24级计算机（留学生）1班	
10	2024331068	MAMMADOV ISRAFIL	24级计算机（留学生）1班	
11	2024331071	AKMALOVA SHAKHRIBONU	24级计算机（留学生）1班	
12	2024331090	PANAH HUMAIRA	24级计算机（留学生）1班	
13	2024331091	TAIFOUR MAHFOUDH	24级计算机（留学生）1班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
14	2024331093	LOTFI OTHMAN	24级计算机（留学生）1班	A2
15	2024331095	ENKHTUR YALALT	24级计算机（留学生）1班	
16	2024331104	RABBI MD ASHIK RAHMAN	24级计算机（留学生）1班	
17	2024331105	FERRAA AHMED	24级计算机（留学生）1班	
18	2024331106	RASENDRIYA MOHAMMED AVICIENNA	24级计算机（留学生）1班	
19	2024331116	OIMSI OUIJDAN	24级计算机（留学生）1班	
20	2024331126	SHAMATOV ALEN	24级计算机（留学生）1班	
21	2024331127	EL MAZILY KHOULOUD	24级计算机（留学生）1班	
22	2024331129	ABU OBAYDA PATHAN REZEN	24级计算机（留学生）1班	
23	2024331134	UNDRAKHBAYAR YESUI	24级计算机（留学生）1班	
24	2024331139	MEZIANE NASR ALLAH	24级计算机（留学生）1班	
25	2024331140	STROGALIN GLEB	24级计算机（留学生）1班	
26	2024331076	EL BYAR OUSSAMA	24级计算机（留学生）1班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
27	2024331149	AL-GABRI AHMED YASEEN HAMOOD	24级计算机（留学生）1班	A3
28	2024331151	LAWRENCE SKY ALVA	24级计算机（留学生）1班	
29	2024331158	TEDJA FICO NATHANEO	24级计算机（留学生）1班	
30	2024331161	ADNAN MD ISTIAQUE ALAM	24级计算机（留学生）1班	
31	2024331169	MAHBOUB SALMA	24级计算机（留学生）1班	
32	2024331177	MEJJAD FATIMA ZAHRA	24级计算机（留学生）1班	
33	2024331179	AHMAD MUHAMMAD	24级计算机（留学生）1班	
34	2024331192	HAMOOD MOHAMMED SEDEQ ALI	24级计算机（留学生）1班	
35	2024331193	MUNKHJARGAL ZOLBAYAR	24级计算机（留学生）1班	
36	2024331202	SOHIBOV MUHAMMAD	24级计算机（留学生）1班	
37	2024331204	KHOMSI OMAR	24级计算机（留学生）1班	
38	2024331401	VEMA AMARA ANASTASIA	24级计算机（留学生）1班	
39	2024331028	ABDULQADER HUSAM ABDULWAOOD ABDULWAHID	24级计算机（留学生）2班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
40	2024331004	SAMMARI ABDESSAMAD	24级计算机（留学生）2班	
41	2024331005	AYOUB DAHMANI IDRISSI	24级计算机（留学生）2班	
42	2024331011	MOGALLI MOHAMMED AHMED QAID MABKHOT	24级计算机（留学生）2班	
43	2024331021	JOUDAR BASMA	24级计算机（留学生）2班	
44	2024331034	FERRAA SALAH EDDINE	24级计算机（留学生）2班	
45	2024331049	MARASINGHE MUDIYANSELAGE AMAYA INDUWAREE DISSANAYAKE	24级计算机（留学生）2班	
46	2024331051	JBARA SAAD	24级计算机（留学生）2班	A4
47	2024331052	JOCELYN MARDI KHANG	24级计算机（留学生）2班	
48	2024331054	BOLD MUNKH ORSHIKH	24级计算机（留学生）2班	
49	2024331059	MAHATHIR AL IMRAN	24级计算机（留学生）2班	
50	2024331065	AHMAD DANIYAL	24级计算机（留学生）2班	
51	2024331066	CHOBANIAN VLADIMIR	24级计算机（留学生）2班	
52	2024331072	KHALILOVA GUNEL	24级计算机（留学生）2班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
53	2024331079	OMAR MARGULAN	24级计算机（留学生）2班	
54	2024331080	ABDULLAH SAMI TAREQ ABDULBAQI	24级计算机（留学生）2班	
55	2024331084	MOHAMED EL ALAMI	24级计算机（留学生）2班	
56	2024331085	ASSIL AYA	24级计算机（留学生）2班	
57	2024331098	HAMIDI NASSIM	24级计算机（留学生）2班	
58	2024331108	NATANAEL	24级计算机（留学生）2班	
59	2024331113	GANBAATAR TUGULDUR	24级计算机（留学生）2班	A5
60	2024331114	ARAFAT YEASIR	24级计算机（留学生）2班	
61	2024331118	BOUFADDA CHARAF-EDDINE	24级计算机（留学生）2班	
62	2024331128	KASSI KELTOUMA	24级计算机（留学生）2班	
63	2024331131	RUKHSHONA TUKHTAEVA	24级计算机（留学生）2班	
64	2024331136	ANARBAEV URANBEK	24级计算机（留学生）2班	
65	2024331146	NYENDWA FLEVIA	24级计算机（留学生）2班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
66	2024331147	KAROUAZ NAOUFEL	24级计算机（留学生）2班	
67	2024331154	MALTSEV EGOR	24级计算机（留学生）2班	
68	2024331162	TADESSE ALAZAR MASRESHA	24级计算机（留学生）2班	
69	2024331163	MALIK MUHAMMAD ALI ABBAS KHAN	24级计算机（留学生）2班	
70	2024331170	AIT JABA ZAKARIAE	24级计算机（留学生）2班	
71	2024331171	AL-DHAIFANI FARES MOHAMMED ALI	24级计算机（留学生）2班	
72	2024331172	GANBOLD ZOLBOO	24级计算机（留学生）2班	A6
73	2024331181	BAYJANOV RAHMANGELDI	24级计算机（留学生）2班	
74	2024331185	HA MINH THU	24级计算机（留学生）2班	
75	2024331194	JOUMADI ABDESSAMAD	24级计算机（留学生）2班	
76	2024331195	GAN-ERDENE TUGULDUR	24级计算机（留学生）2班	
77	2024331203	GRACIELLA CALLISTA EDWARDSON	24级计算机（留学生）2班	
78	2024331402	TRAN VIET ANH	24级计算机（留学生）2班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
79	2024331001	ALI AHMED MOHAMED ABDELMONEIM MANSOUR IBRAHIM	24级计算机（留学生）3班	
80	2024331007	HERATH MUDIYANSELAGE KAVINDYA DISSANAYAKE	24级计算机（留学生）3班	
81	2024331012	ABOHADI BASEM MOHAMMED ABDULLAH	24级计算机（留学生）3班	
82	2024331019	ZARROU HIBA	24级计算机（留学生）3班	
83	2024331022	CHEREPANOV SAVVA	24级计算机（留学生）3班	
84	2024331024	PRONINA ANGELINA	24级计算机（留学生）3班	
85	2024331025	IQBAL CH MUHAMMAD HASHIR	24级计算机（留学生）3班	
86	2024331033	SAIF ASAAD ABDULGHANI AHMED	24级计算机（留学生）3班	A7
87	2024331040	ALTANGEREL SERGELEN	24级计算机（留学生）3班	
88	2024331044	ELARGOUB HAMZA	24级计算机（留学生）3班	
89	2024331050	IBODULLAEV ASRORBEK	24级计算机（留学生）3班	
90	2024331057	BRANDON MANTOVANNI YUNATA	24级计算机（留学生）3班	
91	2024331058	AIT BENFK HAMZA	24级计算机（留学生）3班	
92	2024331063	MUNKHBAT ERKHEMBAYAR	24级计算机（留学生）3班	
93	2024331067	NAJEM KHOULOUD	24级计算机（留学生）3班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
94	2024331073	MANSURZODA MARUF	24级计算机（留学生）3班	
95	2024331082	MOHSEN ASEEL SALEM ATEF AWADH	24级计算机（留学生）3班	
96	2024331086	MD ABU ARSHAD NAIM	24级计算机（留学生）3班	
97	2024331088	GRICH ABDESSAMAD	24级计算机（留学生）3班	
98	2024331099	AIT SI ALI KENZA	24级计算机（留学生）3班	
99	2024331103	SARANEYAWONGSE PANISA	24级计算机（留学生）3班	
100	2024331109	TRI RIDA	24级计算机（留学生）3班	
101	2024331115	ANWAR MUBASHER	24级计算机（留学生）3班	A8
102	2024331117	SUBHA SAUDA SIDDIKA	24级计算机（留学生）3班	
103	2024331120	BATKHUREL SIILEN	24级计算机（留学生）3班	
104	2024331130	RHLALOU ILIASS	24级计算机（留学生）3班	
105	2024331137	ABDULLAKHAMIDOV MUHAMMAD KODIR	24级计算机（留学生）3班	
106	2024331142	LEE PHILIP BERWYN	24级计算机（留学生）3班	
107	2024331145	JOSEPHINE JACINDA	24级计算机（留学生）3班	
108	2024331150	RABII AYMAN	24级计算机（留学生）3班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
109	2024331155	SAHIL NABIR RIYAN	24级计算机（留学生）3班	
110	2024331160	KMASHI MOTASEM BILLAH MOHAMED SAID	24级计算机（留学生）3班	
111	2024331165	KHATIB NASER	24级计算机（留学生）3班	
112	2024331167	PRISANDI GRACELIA	24级计算机（留学生）3班	
113	2024331173	AL-KHANBASHI MUBARAK AHMED MUBARAK AHMED	24级计算机（留学生）3班	
114	2024331174	EL GASSOUSI YASSINE	24级计算机（留学生）3班	
115	2024331178	MAHTAB ABRAR SHAHRIAR	24级计算机（留学生）3班	
116	2024331182	SUKHBAATAR BAATARKHUU BAATARKHUU	24级计算机（留学生）3班	A9
117	2024331199	SAIR ABDULLAH KHALID	24级计算机（留学生）3班	
118	2024331200	BATSAIKHAN UNDRAL	24级计算机（留学生）3班	
119	2024331201	CONG HOANG LAN	24级计算机（留学生）3班	
120	2024331208	ASLAN AISHA	24级计算机（留学生）3班	
121	2024331211	HALIM KEVIN NATANIEL	24级计算机（留学生）3班	
122	2024331406	NZALI FEZEU YURIY BRYAN	24级计算机（留学生）3班	
123	2024331100	BIASSARILA BONI JUNIOR	24级计算机（留学生）3班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
124	2024331003	AIT MANNA IMANE	24级计算机（留学生）4班	
125	2024331008	WIJESINGHE MUDIYANSELAGE DULMINI IMASHA WIJESINGHE	24级计算机（留学生）4班	
126	2024331009	G HANEM MOHAMMED AMGAD SAEED	24级计算机（留学生）4班	
127	2024331013	AL-BADWI EZZAT ABDULKAREEM GHALLAB OTHMAN	24级计算机（留学生）4班	
128	2024331018	HAMZAOUI SOULAYMAN	24级计算机（留学生）4班	
129	2024331027	MADANI ALAOUI YOUNESS	24级计算机（留学生）4班	
130	2024331035	NAURYZBAY RASSUL	24级计算机（留学生）4班	
131	2024331041	ALI DHAIF ALLAH SADEQ ALI	24级计算机（留学生）4班	A10
132	2024331042	BATJARGAL DULGUUN	24级计算机（留学生）4班	
133	2024331045	EL AMRANI SALEM ABDERRAHMAN	24级计算机（留学生）4班	
134	2024331056	ARWA AHMED HAFEZ MOHAMED ElSAYED	24级计算机（留学生）4班	
135	2024331064	BATTUR ODMICHID	24级计算机（留学生）4班	
136	2024331070	KHALIMOV BEKHRUZBEK	24级计算机（留学生）4班	
137	2024331074	KASYMOV ISMAIL	24级计算机（留学生）4班	
138	2024331077	MAHRACH ANAS	24级计算机（留学生）4班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
139	2024331078	MSIGWA AMOSI EXAVERY	24级计算机（留学生）4班	A11
140	2024331081	ISHAQ AHTISHAM	24级计算机（留学生）4班	
141	2024331083	MOQBEL OSAMAH ABDULAZIZ AHMED	24级计算机（留学生）4班	
142	2024331087	GALIMOV DINAR	24级计算机（留学生）4班	
143	2024331089	ZOUAOUI HATIM	24级计算机（留学生）4班	
144	2024331092	JORDAN YEO HUNG EN	24级计算机（留学生）4班	
145	2024331094	ASHRAF AHONA	24级计算机（留学生）4班	
146	2024331096	SETIAWAN VIRGIL CONSTANTINE	24级计算机（留学生）4班	
147	2024331102	KONURBAEVA ERIKA	24级计算机（留学生）4班	
148	2024331119	SYED REFAT AL SAMI	24级计算机（留学生）4班	
149	2024331123	BADACH WALID	24级计算机（留学生）4班	
150	2024331124	NGUYEN HONG PHUONG	24级计算机（留学生）4班	
151	2024331132	MOUHIDDINE LOUKMANE	24级计算机（留学生）4班	
152	2024331133	VOLOVOI PAVEL	24级计算机（留学生）4班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
153	2024331143	MBAO KELVIN JAMES	24级计算机（留学生）4班	A12
154	2024331153	AHMADI YAZDI DZUNURRAYN	24级计算机（留学生）4班	
155	2024331156	REHMAN MUHAMMAD ABDUL	24级计算机（留学生）4班	
156	2024331157	FELICIA	24级计算机（留学生）4班	
157	2024331176	MOUZOUN YOUNESS	24级计算机（留学生）4班	
158	2024331183	DEKIMECHE ABDERRAHIM	24级计算机（留学生）4班	
159	2024331187	BATSARAN GANTUSHIG	24级计算机（留学生）4班	
160	2024331190	SIASSE ABDELHAMID	24级计算机（留学生）4班	
161	2024331198	NADIR REDA	24级计算机（留学生）4班	
162	2024331205	BAJWA MOAZ NADEEM	24级计算机（留学生）4班	
163	2024331206	BADAR-UUGAN BATTUSHIG	24级计算机（留学生）4班	
164	2024331207	KHAKIMJONOV ASADBEK	24级计算机（留学生）4班	
165	2024331403	KENNETH PUTRA RIYANTO	24级计算机（留学生）4班	
166	2024331404	MSANGI ABUBAKAR ABEID	24级计算机（留学生）4班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
167	2024332002	WANDERA DAVIS BLESSINGS	24级机器人工程（留学生）1班	
168	2024332003	NILOY MD SAHIL SADIK	24级机器人工程（留学生）1班	
169	2024332004	ABDUL MOIZ	24级机器人工程（留学生）1班	
170	2024332005	AL-AGHBARI AL-BARAA WAEL ASAAD SHAHER	24级机器人工程（留学生）1班	
171	2024332006	ALACHAHER HALA MANSOURIA	24级机器人工程（留学生）1班	
172	2024332008	EL-ATTARI DOUAE	24级机器人工程（留学生）1班	
173	2024332009	UVAYDULLAEV OTABEK	24级机器人工程（留学生）1班	A13
174	2024332010	ARIS IBRAHIM ZABIDI BIN AZMIL	24级机器人工程（留学生）1班	
175	2024332011	AKASH MD AMDADUL HOQUE	24级机器人工程（留学生）1班	
176	2024332012	RAVSHANOV AZIZBEK	24级机器人工程（留学生）1班	
177	2024332014	TAY YU ZHE	24级机器人工程（留学生）1班	
178	2024332015	KACHALIN KIRILL	24级机器人工程（留学生）1班	
179	2024332016	KOTTAPOLA SAMARANAYAKALAGE KAVINDU SHELUM	24级机器人工程（留学生）1班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
180	2024332017	TAWANSANGARAM PHAWATH	24级机器人工程（留学生）1班	A14
181	2024332018	NAIM ABDULLAH	24级机器人工程（留学生）1班	
182	2024332019	SAJIDE MARWA	24级机器人工程（留学生）1班	
183	2024332020	AL-EZZI ABDULLAH FATEHI SALEH HUSSEIN	24级机器人工程（留学生）1班	
184	2024332022	BETANCUR ROJAS DARICK JOHANNES	24级机器人工程（留学生）1班	
185	2024332023	A RAHIM MD	24级机器人工程（留学生）1班	
186	2024332024	TADEUS FABIO SAMOSIR	24级机器人工程（留学生）1班	
187	2024332025	CHEKAOUI HIBA	24级机器人工程（留学生）1班	
188	2024332027	FREDERICK NOBLE UCHENNA	24级机器人工程（留学生）1班	
189	2024332028	DEBBAB MERYEM	24级机器人工程（留学生）1班	
190	2024332029	BARALANGUI CHERUBIN LUDEN	24级机器人工程（留学生）1班	
191	2024332030	OUKHADJOU MARWA	24级机器人工程（留学生）1班	
192	2024332034	LOPEZ LOPEZ DONAL DIXON	24级机器人工程（留学生）1班	
193	2024332001	BAYAD ZAKARIA	24级机器人工程（留学生）1班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
194	2024332202	MOHAMED WAEIL KHALIL ABOUESMAIL	24级电气工程及其自动化（留学生）1班	A15
195	2024332204	LEE KHAI YUEH	24级电气工程及其自动化（留学生）1班	
196	2024332206	ZAYD ABDULLAH	24级电气工程及其自动化（留学生）1班	
197	2024332207	ERGASHEV JAHONGIR	24级电气工程及其自动化（留学生）1班	
198	2024332209	ALAM MOSTAKENUL	24级电气工程及其自动化（留学生）1班	
199	2024332210	KAHUMULA MUMBERE DANNY	24级电气工程及其自动化（留学生）1班	
200	2024332211	HANI NIZAR	24级电气工程及其自动化（留学生）1班	
201	2024332213	TAKOUK SOFIANE	24级电气工程及其自动化（留学生）1班	
202	2024332214	AL-BAHR AL-YMAAN FUAD ALI MOHAMMED	24级电气工程及其自动化（留学生）1班	
203	2024332215	UDDIN MD SABUJ	24级电气工程及其自动化（留学生）1班	
204	2024332217	ABDRAHMAN MAHAMAT NOUR ABAKAR AHMAT	24级电气工程及其自动化（留学生）1班	
205	2024332218	GIYOZOV MIRZOALI	24级电气工程及其自动化（留学生）1班	
206	2024332219	AIT TALEB ACHRAF	24级电气工程及其自动化（留学生）1班	
207	2024332220	ELSADIG ELEMAM GOUDA MOHAMED	24级电气工程及其自动化（留学生）1班	

4. Course arrangements - Group A1 to Group A16

NO.	Student ID	NAME	CLASS	GROUP
208	2024332221	BELKAMCH MOUAD	24级电气工程及其自动化（留学生）1班	
209	2024332222	OTGONDORJ PUREVDORJ	24级电气工程及其自动化（留学生）1班	
210	2024332223	THUTA NAING	24级电气工程及其自动化（留学生）1班	
211	2024332227	CHENG ILGIZ	24级电气工程及其自动化（留学生）1班	
212	2024332228	TOURBIH YOUSSEF	24级电气工程及其自动化（留学生）1班	
213	2024332231	UDDIN MD MESHBA	24级电气工程及其自动化（留学生）1班	
214	2024332232	SHAYAN MUNIR	24级电气工程及其自动化（留学生）1班	
215	2024332234	UWIMPUHWE TAMARA NANIS	24级电气工程及其自动化（留学生）1班	A16
216	2024332235	HUSSAIN ABDUR RAHMAN	24级电气工程及其自动化（留学生）1班	
217	2024332237	ALWAN ABDULLAH MOHAMMED ABDULLAH	24级电气工程及其自动化（留学生）1班	
218	2024332241	ADIL RIDWAN BASHAR	24级电气工程及其自动化（留学生）1班	
219	2024332242	PENG SOPANHAROTH	24级电气工程及其自动化（留学生）1班	
220	2024332243	MOHAMED ABDULBARI AHMED	24级电气工程及其自动化（留学生）1班	
221	2024332205	EL ACHIR SAAD	24级电气工程及其自动化（留学生）1班	
222	2024332224	DIYOUBA DAMTA NADINE REINE	24级电气工程及其自动化（留学生）1班	

5. Course requirements

Instruction booklet

Instruction Book of University Physics Experiment A

Table of Contents

Experiment I: Michelson Interferometer 1
Experiment II: Young's Modulus of a Metallic Wire 12
Experiment III: Collision and Targeting Experiment 17
Experiment IV: The Hall Effect 21
Experiment V: Liquid Surface Tension Coefficient Measurement 25
Experiment VI: Characteristics of RLC circuits 32
Experiment VII: Speed of Sound in Air 39

Each experiment:

1. Pre-lab
2. Objectives and Tasks
3. Principles
4. Set-up
5. Procedure
6. Precautions
7. Lab-report requirements

5. Course requirements

Lab-reports

Each experiment:	1. Pre-lab assignment	(2 marks)
	2. Data acquisition	(3 marks)
	3. Data processing	(3 marks)
	4. Data analysis	(1 marks)
	5. Discussion	(1 marks)

Total: 10 marks

As a prerequisite for each experiment, the pre-lab assignment must be completed in advance and then signed and marked by the lecturer in the begining of the laboratory class, for otherwise you will *not* be allowed to conduct your experiment.

5. Course requirements

Lab-reports

For each experiment:

1. Complete your pre-lab assignment and lab report using the template we provide.

2. Submit your lab report in pdf, within ONE week after your laboratory class (Submission link to be provided via the QQ group):

3. When submitting, fill in your name and student ID as well as the name of the experiment according to the system's requirements.

5. Course requirements

Lab-reports

3. Record the raw data on a separate sheet, and let the lecturer check your data and sign on the sheet.

4. Process your raw data right after the experiment, and present the corresponding details of data-processing in your lab report.

5. Data processing includes step-by-step calculations, estimations of errors and uncertainties, and graphical presentations of results.

6. Answer the questions in the dicussion section using existing knowledge of physics.

5. Course requirements

Laboratory discipline

For each experiment

1. You should arrive at the laboratory on time, ideally 15 minutes in advance. If you are late for more than 10 minutes, you will NOT be allowed to attend the laboratory class.

2. Upon arrival at the laboratory, take attendance, pick one set of equipment for your experiment, and let the lecturer mark your pre-lab assignment.

3. Upon completion of your experiment, tidy up your area before leaving the laboratory.

5. Course requirements

Laboratory discipline

4. Pay special attention to the precautions and safety measures advised by the lecturer.

5. Plagarism of lab reports and fabrication of experimental data are strictly prohibited. Once discovered, the corresponding lab report will be disqualified with a zero score, and you will fail the whole course.

6. If you cannot attend the laboratory class as scheduled, you can reschedule your slot in accordance with our course schedule.

7. To pass this course, you must have completed all 7 experiments and obtained an overall score of at least 60% of the total available marks.

6. Errors and uncertainties in measurements

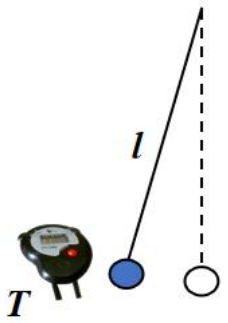
Measurements

A measurement is the comparison of an unknown quantity with a standard quantity (i.e. a unit).

A direct measurement is the direct comparison of an unknown quantity with a standard quantity, through the use of an instrument.

An indirect measurement is the evaluation of an unknown quantity through its mathematical relation with directly measured quantities. For example,

$g = (4\pi^2 l)/T^2$



6. Errors and uncertainties in measurements

Measured value A

True value A_0

Absolute error $\Delta A \equiv A - A_0$

Relative error $E \equiv (\Delta A/A_0) \times 100\%$

If the true value is approximated by the average value $\overline{A}$, we have

Absolute deviation $\Delta A \equiv A - \overline{A}$

Percentage error $E \equiv (\Delta A/\overline{A}) \times 100\%$

6. Errors and uncertainties in measurements

Systematic errors and random errors

Systematic errors, which arise from instrumental bias, environmental interference, procedural mistakes, or human errors, affect measurements in a uniform direction or manner, leading to biased results. Such errors can be minimised via an improved method of measurement, a properly modified empirical relation, or a proper calibration / set-zero of the instrument.

Random errors, which occur by chance or due to some unpredictable factors, affect measurements in a random manner. Such errors can be minimised by conducting a sufficiently large number of measurements such that the errors approach a normal distribution.

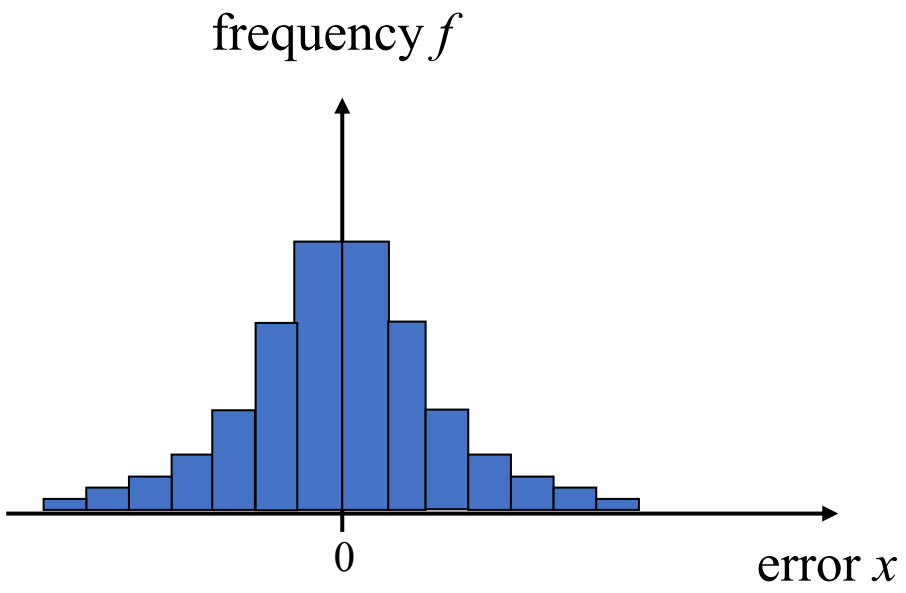
6. Errors and uncertainties in measurements

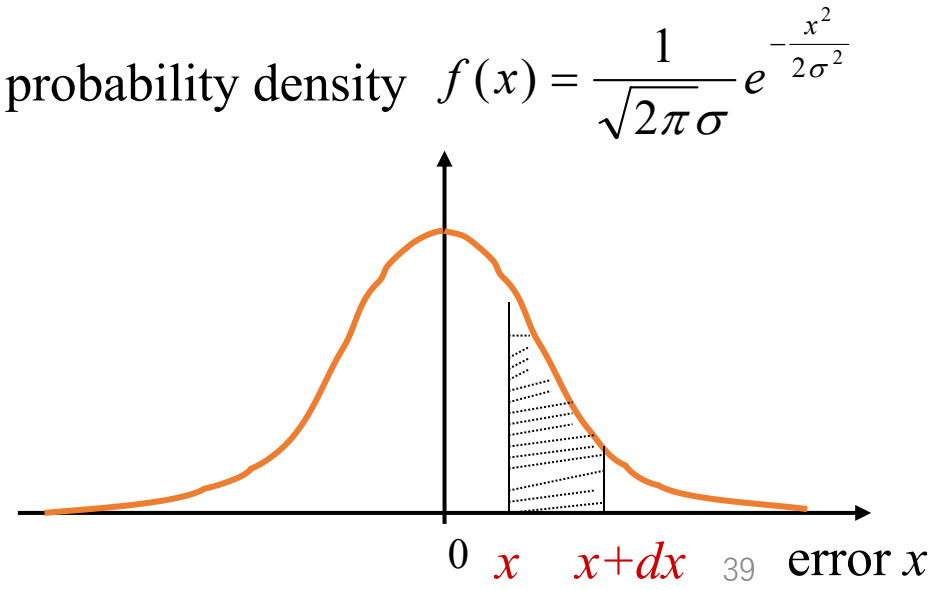
Distribution of random errors

For the ith measurement:

Random error $x_i = A_i - A_0$

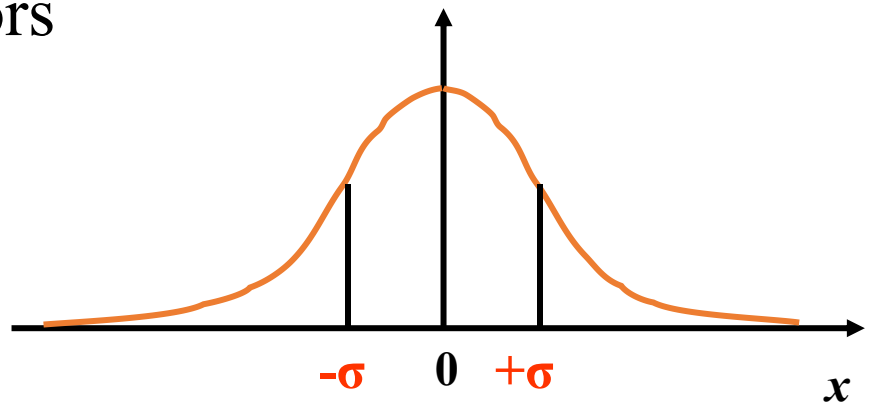
Approaching a normal distribution in the limit of infinite measurements:





6. Errors and uncertainties in measurements

Distribution of random errors

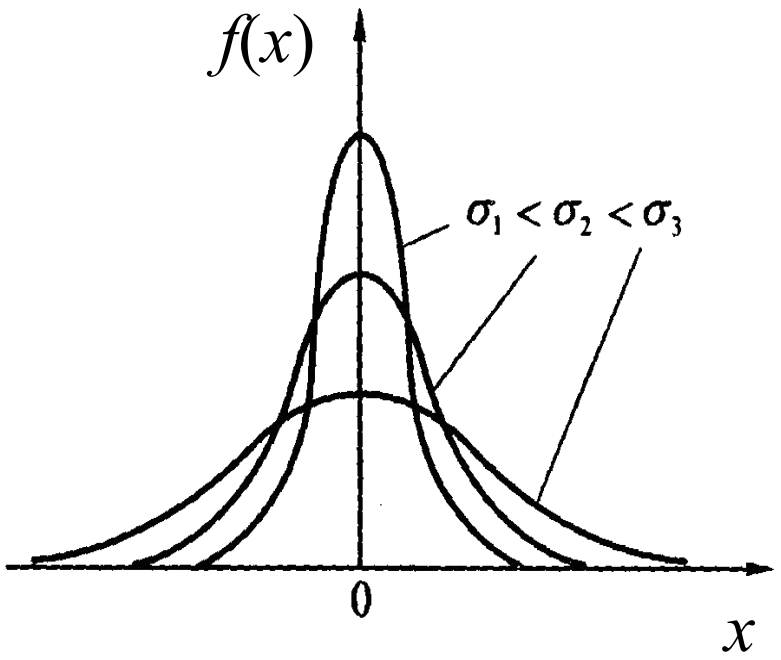


Standard error $\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(A_i - A_0)^2}$

Confidence interval $[-\sigma,\sigma]$

Coverage probability / confidence level $P_{x\in[-\sigma,\sigma]} = \int_{-\sigma}^{\sigma} f(x)dx = 0.683$

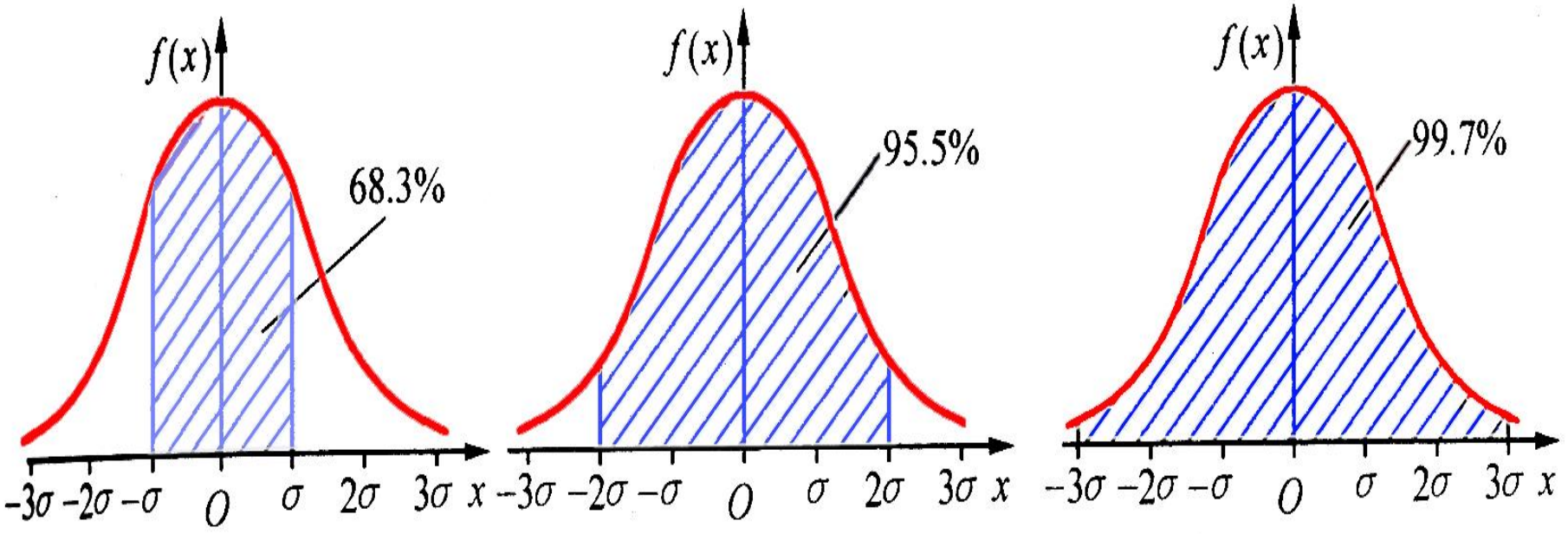
The greater the value of σ, the greater the the probability for the occurrance of large errors



6. Errors and uncertainties in measurements

Distribution of random errors

Confidence levels at various confidence intervals:



Assumptions: (1) Infinite number of measurements (which cannot be achieved)
(2) The true value A_0 is known (which is not the case)

6. Errors and uncertainties in measurements

Distribution of random errors

Approximation of the standard error $\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(A_i - A_0)^2}$

for an infinite number of measurements and for an unknown value A_0

by the standard deviation $S = \sqrt{\frac{\sum_{i=1}^{n}(A_i - \overline{A})^2}{n-1}}$

for a finite number of measurements and for the corresponding average $\overline{A}$

(Bessel's correction: use of $n - 1$ instead of n)

6. Errors and uncertainties in measurements

Distribution of random errors

The average $\overline{A}$, which is obtained from a finite number of measurements, also exhibits random absolute deviations x’ about an average value.

Its standard deviation is given by

$$S' = \frac{S}{\sqrt{n}} = \sqrt{\frac{\sum_{i=1}^{n} (A_i - \overline{A})^2}{n(n-1)}}$$

Confidence level $P_{x' \in [-S', S']} = \int_{-S'}^{S'} f(x')dx' = 0.683$

6. Errors and uncertainties in measurements

Distribution of random errors

An uncertainty U refers to the numerical range $\overline{A} \pm U$ which a measured result falls within at a particular confidence level. Here we choose this probability to be equal to $P = 0.683$.

A type-A uncertainty is obtained from the statistical analysis of a series of measurements. For $P = 0.683$, it is given by

$$S_i \equiv S' = \sqrt{\frac{\sum_{i=1}^{n} (A_i - \overline{A})^2}{n(n-1)}}$$

6. Errors and uncertainties in measurements

Distribution of random errors

A type-B uncertainty, denoted here as S_j, is obtained by non-statistical means. Here we only consider type-B uncertainties due to the presence of instrumental errors $\Delta_{\text{instrument}}$:

$$S_j = \frac{\Delta_{\text{instrument}}}{C}, \quad \begin{cases} C = \sqrt{3}, \text{ uniform distribution of errors} \\ C = 3, \text{ normal distribution of errors} \end{cases}$$

Total uncertainty $U = \sqrt{S_i^{\,2} + S_j^{\,2}}$

6. Errors and uncertainties in measurements

Precision and accuracy

Accuracy refers to how close the measured results of an experiment are to the true value.

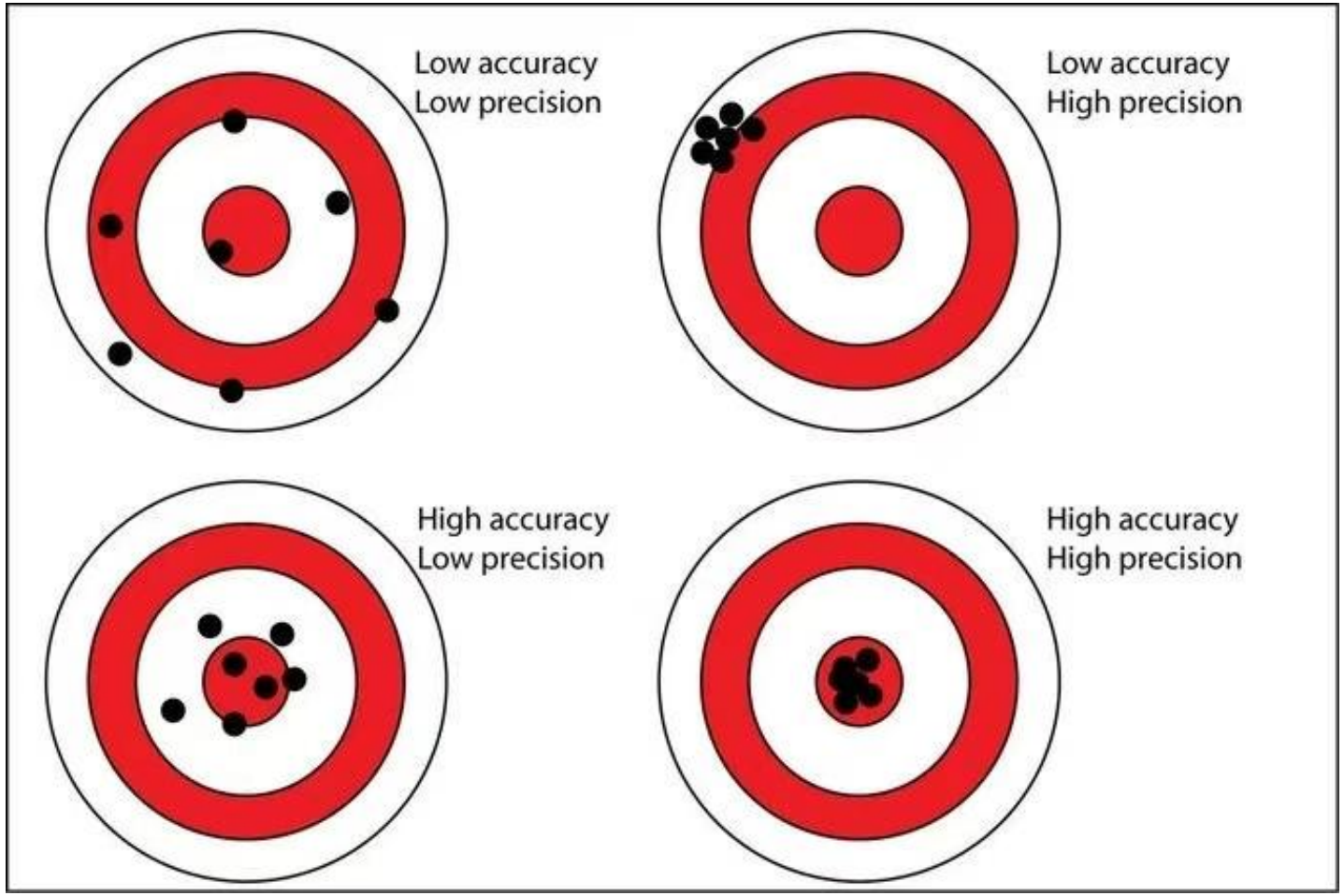
Precision refers to how close the measured results of an experiment are to one another.

A larger value of the standard error σ corresponds to a wider spread of data due to random errors and hence a lower precision.

Any systematic error shifts the whole set of data away from the true value, thus lowering the accuracy of the measurements.

6. Errors and uncertainties in measurements

Precision and accuracy



6. Errors and uncertainties in measurements

Presentation of experimental results

Results from a direct measurement: $A = \overline{A} \pm U$

$$\text{Percentage error } E = \frac{U}{\overline{A}} \times 100\,\%$$

$$\text{Confidence level } P = 0.683$$

Results from an indirect measurement:

Let B be an indirectly measured quantity that is related to the directly measured quantities $x_1, x_2, \cdots, x_m$ as follows:

$$B = f(x_1, x_2, x_3, \ldots, x_m)$$

6. Errors and uncertainties in measurements

Presentation of experimental results

Results from an indirect measurement: $B = \overline{B} \pm U$

Percentage error $E = \dfrac{U}{\overline{B}} \times 100\,\%$

Confidence level $P = 0.683$

where $\overline{B} = f(\overline{x}_1, \overline{x}_2, \overline{x}_3, \ldots, \overline{x}_m)$

For $x_i = \overline{x}_i \pm U_i \quad i \in [1, m]$, the overall uncertainty for B is given by

$$U = \sqrt{\left(\frac{\partial f}{\partial x_1}\right)^2 U_1^2 + \left(\frac{\partial f}{\partial x_2}\right)^2 U_2^2 + \cdots + \left(\frac{\partial f}{\partial x_m}\right)^2 U_m^2} = \overline{B}\sqrt{\left(\frac{\partial \ln f}{\partial x_1}\right)^2 U_1^2 + \left(\frac{\partial \ln f}{\partial x_2}\right)^2 U_2^2 + \cdots + \left(\frac{\partial \ln f}{\partial x_m}\right)^2 U_m^2}$$

7. Significant figures of measured quantities

The significant figures of a measured quantity refer to the set of digits required to express the quantity to the required degree of accuracy, starting from the leftmost non-zero digit and ending with a digit relevant to the resolution limit of the measurement. Here we use the following rules to determine the significant figures in a measured quantity:

1. Any non-zero digit is a significant figure. For example, the quantity 4.321 kg consists of four significant figures.

2. Any zero between two non-zero digits is a significant figure. For example, the quantity 90.05 kg consists of four significant figures.

7. Significant figures of measured quantities

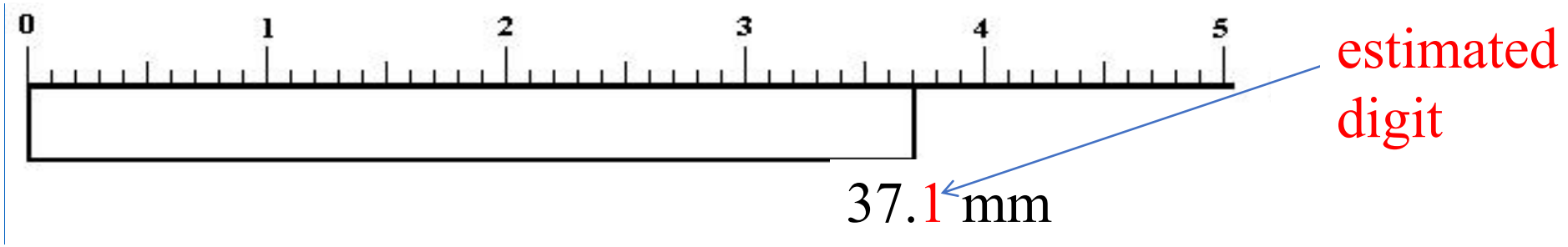
3. Any zero to the left of the first non-zero digit is *not* a significant figure. For example, the quantity 0.015 kg consists of two significant figures.

4. Any zero to the right of the rightmost non-zero digit is a significant figure. For example, the quantity 0.0450 kg consists of three significant figures.

5. Any zero to the right of the rightmost non-zero digit in an integer may or may not be significant, depending on the resolution of the measurement. For example, the quantity 10500 km consists of three significant figures if the resolution of measurement is 100 km, and it consists of five significant figures if the resolution of measurement is 1km.

7. Significant figures of measured quantities

6. Any directly measured quantity should be expressed up to the rightmost significant digit of the instrumental error, with the rightmost digit being an estimated value. For a ruler with a resolution of 1 mm, the instrumental error is about half of this resolution, i.e. around 0.5 mm, so that any measured length should be expressed up to the first decimal place:



7. For a voltmeter or an ammeter, the instrumental error is typically 0.5% of the chosen scale. For example, if the chosen scale of a voltmeter is 10 V, the instrumental error is around 0.05 V, and the measured voltage should be expressed up to the second decimal place.

7. Significant figures of measured quantities

8. For any measured quantity, the number of significant figures is independent of the position of the decimal point. For example, the quantity 10.50 mm can equivalently be written as 1.050 cm (but *not* 10500000 nm).

9. The uncertainty of any measured quantity should be expressed with only one or two significant figures. For example, if the computed value of an uncertainty is 0.05106, it should be rounded up to a single significant figure as 0.06. If the leftmost significant figure is 1 or 2, e.g. 0.02135, it should be rounded up to two significant figures, e.g. 0.022.

10. The calculated value of any indirectly measured quantity should be rounded off to the rightmost significant digit of the corresponding uncertainty, according to the following rules for rounding:

7. Significant figures of measured quantities

If the digit one step to the right from the rounding digit is 0, 1, 2, 3, or 4, round down. For example, the number 5.76433 is rounded to four significant figures as 5.764.

If the digit one step right from the rounding digit is 6, 7, 8, or 9, round up. For example, the number 5.76473 is rounded to four significant figures as 5.765.

If the rounding digit is an odd number and the digit one step right from the rounding digit is 5, round up. For example, the number 5.76353 is rounded to four significant figures as 5.764.

If the rounding digit is an even number and the digit one step right from the rounding digit is 5, keep the rounding digit unchanged. For example, the number 5.76453 is rounded to four significant figures as 5.764.

7. Significant figures of measured quantities

In general, the accuracy of a calculated result is limited by the least accurate measurement involved in the calculation.

In addition and subtraction, the result should be rounded off to the rightmost common significant digit among all components. For example, 100 + 23.643 = 123.643, which should be rounded to 124.

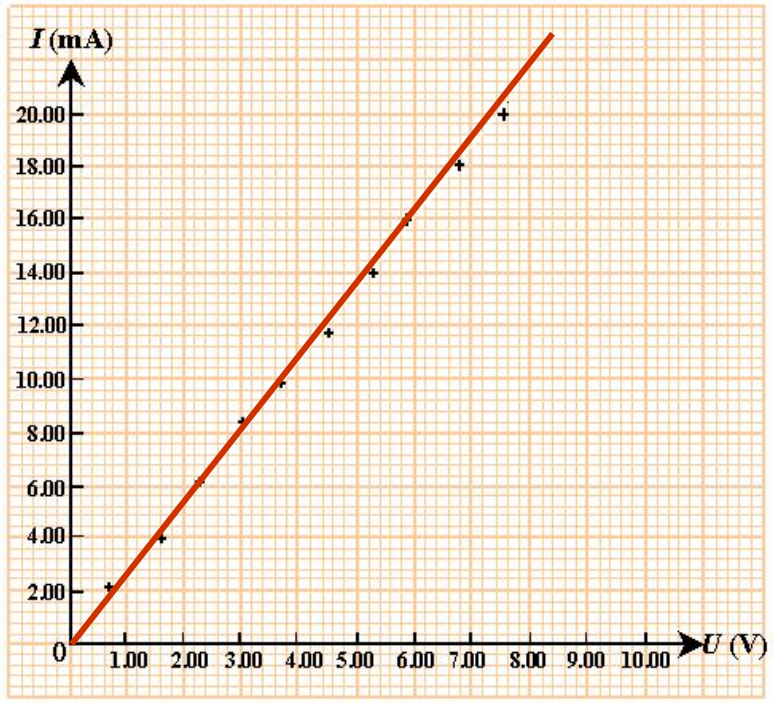
In multiplication and division, the result should be rounded off so as to have the same number of significant figures as in the component with the least number of significant figures. For example, 3.0 (with two significant figures) × 12.60 (with four significant figures) = 37.8 (with three significant figures), which should be rounded off to 38 (with two significant figures).

8. Methods of data analysis

1. Tabulation of data

$U\,(V)$	0.74	1.52	2.33	3.08	3.66	4.49	5.24	5.98	6.76	7.50
$I\,(mA)$	2.00	4.01	6.22	8.20	9.75	12.00	13.99	15.92	18.00	20.01

2. Graphical representation of data



8. Methods of data analysis

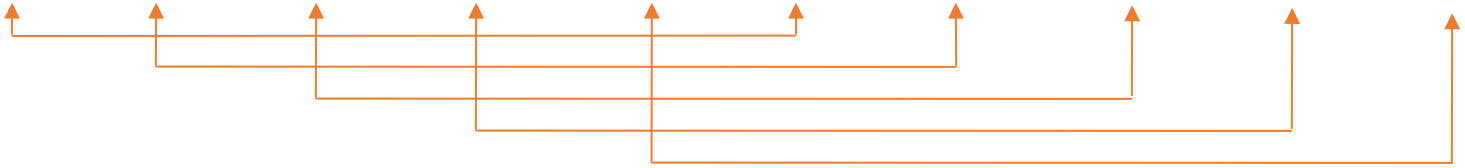
3. Method of successive differences:

Assume $y = kx + b$ with a total of $2N$ data points of x at regular spacings, and assume that the measurement errors of x are negligible. The slope k can be computed from the equation

$$k = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{y_{i+N} - y_i}{x_{i+N} - x_i}\right)$$

Example:

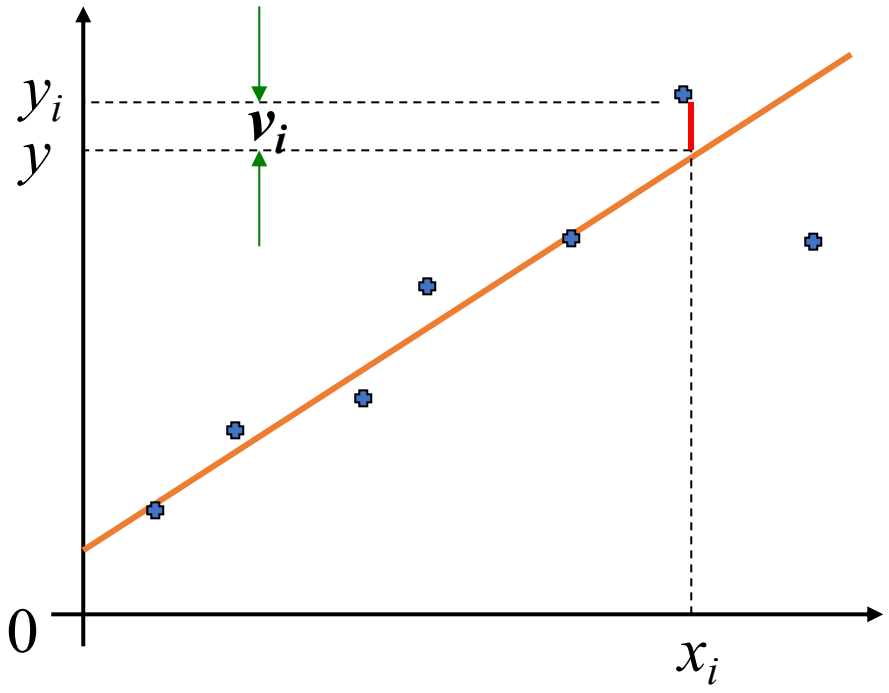
Voltage (V)	**0.00**	**1.00**	**2.00**	**3.00**	**4.00**	**5.00**	**6.00**	**7.00**	**8.00**	**9.00**
Current (mA)	**0.00**	**0.50**	**1.02**	**1.49**	**2.05**	**2.51**	**2.98**	**3.52**	**4.00**	**4.48**



8. Methods of data analysis

4. Method of least squares

Let $y = kx + b$



$$v_i = y_i - y = y_i - (kx_i + b)$$

$$\sum_{i=1}^{n} v_i^2 = \sum_{i=1}^{n} [y_i - (kx_i + b)]^2 = \min$$

$$\Rightarrow \begin{cases} \dfrac{\partial}{\partial k}\left(\sum_{i=1}^{n} v_i^2\right) = 0 \\ \dfrac{\partial}{\partial b}\left(\sum_{i=1}^{n} v_i^2\right) = 0 \end{cases} \Rightarrow \begin{cases} k = \dfrac{\overline{x}\,\overline{y} - \overline{xy}}{\overline{x}^2 - \overline{x^2}} \\ b = \overline{y} - k\overline{x} \end{cases}$$

Correlation coefficient $r = \dfrac{\overline{(xy)} - \overline{(x)}\,\overline{(y)}}{\sqrt{(\overline{x^2} - \overline{x}^2)(\overline{y^2} - \overline{y}^2)}} \in [-1, 1]$

~ End of Lecture ~